

Smart Parking

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Problem

Drivers waste an average of 15–20 minutes searching for parking, generating over 30% of inner-city traffic congestion and unnecessary carbon emissions. Traditional parking facilities only display binary "Full / Open" signs at the entrance, leaving drivers to blindly cruise multi-level garages without floor or spot-level guidance, while physical ticketing booths cause severe bottlenecks and long queues upon exit.

Objectives

- **Spot-Level Guidance:** Direct drivers to exact available spots.
- **Live Space Monitoring:** Track individual spot occupancy in real time.
- **Frictionless Exit:** Fast, contactless mobile payment without exit queues.

The Solution

- **Driver App:** Multi-level spot maps & mobile checkout.
- **Manager Portal:** Real-time occupancy analytics & slot overrides.
- **Real-Time Engine:** sync using WebSockets.
- **Stack:** React 19, Node.js, Express, Socket.io, MongoDB, PayPal.

Results

- **Full End-to-End Flow Proven:** Successfully demonstrated the complete journey in testing: a driver discovers a free spot → parks → pays via PayPal → receives an email receipt → spot status updates automatically.
- **Live Multi-User Sync:** Verified that when a spot's status changes, it updates across multiple open browser screens simultaneously with zero page refreshes.
- **Simulation Test Passed:** The automated simulation engine successfully ran continuous randomized parking cycles without crashing the server or corrupting database records.

Impact & Future

- **Industry impact:** Low-cost IoT model; eliminates ticket machines & queues.
- **Future Work:** AI availability forecast in lot status.

Flow

